

EC-390: Digital Signal Processing

LECTURE NO 13: IIR (Infinite Impulse Response) Filters

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Organization of Lecture

1. Introduction.
2. Impulse Invariant Transformation.
3. Bilinear Transformation.

1. Introduction

The specification of a digital filter will be desired frequency response, $H_d(e^{j\omega})$. The desired impulse response, $h_d(n)$ of the digital filter can be obtained by taking inverse Fourier transform of $H_d(e^{j\omega})$. Now, the $h_d(n)$ will be an infinite duration discrete time signal defined for all values of n in the range $-\infty$ to $+\infty$. The filters designed by considering all the infinite samples of impulse response are called IIR (*Infinite Impulse Response*) filters.

In digital domain, the processing of infinite samples of impulse response is practically not possible. Hence direct design of IIR filter is not possible. Therefore, the IIR filters are designed via analog filters.

In design of IIR filter, the specification of an IIR filter is transformed to specification of an analog filter and an analog filter with transfer function, $H(s)$ is designed to satisfy the specification. Then the analog filter is transformed to digital filter with transfer function, $H(z)$.

The popular transformation techniques used for transforming analog filter transfer function $H(s)$ to digital filter transfer function $H(z)$ are bilinear and impulse.

The frequency response $H(e^{j\omega})$ of the digital filter can be obtained by letting $z = e^{j\omega}$ in the transfer function $H(z)$ of the filter.

$$H(s) \xrightarrow{\text{Inverse LT}} h(t) \xrightarrow{t = nT} h(nT) \xrightarrow{h(n)} h(n) \xrightarrow{\text{Z-Transform}} H(z) = Y(z)/X(z).$$

1. Introduction

Advantages of Digital Filters

1. The values of resistors, capacitors and inductors used in the analog filters changes with temperature. Since digital filters do not have these components, they have high thermal stability.
2. In digital filters the precision of the filter depends on the length (or size) of the registers used to store the filter coefficients. Hence by increasing the register bit-length (in hardware) the performance characteristics of the filter like accuracy, dynamic range, stability and frequency response tolerance, can be enhanced.
3. The digital filters are programmable. Hence the filter coefficients can be changed at any time to implement adaptive features.
4. A single filter can be used to process multiple signals by using the techniques of multiplexing.

1. Introduction

Disadvantages of Digital Filters

1. The bandwidth of the discrete signal is limited by the sampling frequency. The bandwidth of real discrete signal is half the sampling frequency. $\rightarrow F_s \geq 2F_{max}$
2. The performance of the digital filter depends on the hardware (i.e., depends on the bit length of the registers in the hardware) used to implement the filter.

Important Features of IIR Filters

1. The physically realizable IIR filters do not have linear phase.
2. The IIR filter specifications include the desired characteristics for the magnitude response only.

1. Introduction

Digital Filter	Analog Filter
1. Operates on digital samples(or sampled version) of the signal.	1. Operates on analog signals(or actual signals).
2. It is governed (or defined) by linear difference equation.	2. It is governed (or defined) by linear differential equation.
3. It consists of adders, multipliers and delays implemented in digital logic(either in hardware or software or both).	3. It consists of electrical components like resistors, capacitors and inductors.
4. In digital filters the filter coefficients are designed to satisfy the desired frequency response.	4. In analog filters the approximation problem is solved to satisfy the desired frequency response.

Table 1 : Comparison of Digital and Analog Filters

2. Impulse Invariant Transformation

The main idea behind this technique is to preserve the frequency response characteristics of the analog filter. It can be stated that the frequency response of digital filter will be identical with the frequency response of the corresponding analog filter if the sampling time period T is selected sufficiently small (or the sampling frequency should be high) to minimize (or avoid completely) the effects of aliasing.

Let, $h(t)$ = Impulse response of analog filter

The Laplace transform of the analog impulse response $h(t)$ gives the transfer function of analog filter.

$\therefore$ Transfer function of analog filter, $H(s) = \mathcal{L}\{h(t)\}$.

- $T = \text{Sampling Time Period} = 1 / F_s$
- $F_s = \text{Sampling Frequency} = 1 / T$

2. Impulse Invariant Transformation

When $H(s)$ has N number of distinct poles, it can be expressed as shown in equation (2.1) by partial fraction expansion.

$$H(s) = \sum_{i=1}^N \frac{A_i}{s + p_i} = \frac{A_1}{s + p_1} + \frac{A_2}{s + p_2} + \dots + \frac{A_N}{s + p_N} \dots(2.1)$$

On taking inverse Laplace transform of equation (2.1) we get,

$$\mathcal{L}\{e^{-at} u(t)\} = \frac{1}{s + a}$$

$$h(t) = \sum_{i=1}^N A_i e^{-p_i t} u(t) = A_1 e^{-p_1 t} u(t) + A_2 e^{-p_2 t} u(t) + \dots + A_N e^{-p_N t} u(t) \dots(2.2)$$

where, $u(t)$ = Continuous time unit step function.

2. Impulse Invariant Transformation

Let, T = Sampling period.

$h(n)$ = Impulse response of digital filter.

The impulse response of the digital filter is obtained by uniformly sampling the impulse response of the analog filter

$$\therefore h(n) = h(t) \Big|_{t=nT} = h(nT)$$

Therefore the impulse response $h(n)$ can be obtained from equation (2.2) by replacing t by nT .

$$\begin{aligned} \therefore h(n) &= h(t) \Big|_{t=nT} = h(nT) = \sum_{i=1}^N A_i e^{-p_i nT} u(nT) \\ &= A_1 e^{-p_1 nT} u(nT) + A_2 e^{-p_2 nT} u(nT) + \dots + A_N e^{-p_N nT} u(nT) \dots (2.3) \end{aligned}$$

On taking $\mathcal{Z}$ -transform of equation (2.3) we get,

$$\begin{aligned} H(z) = \mathcal{Z}\{h(n)\} &= A_1 \frac{1}{1 - e^{-p_1 T} z^{-1}} + A_2 \frac{1}{1 - e^{-p_2 T} z^{-1}} + \dots \\ &+ A_N \frac{1}{1 - e^{-p_N T} z^{-1}} = \sum_{i=1}^N A_i \frac{1}{1 - e^{-p_i T} z^{-1}} \dots (2.4) \end{aligned}$$

$$\boxed{\mathcal{Z}\{e^{-anT} u(nT)\} = \frac{1}{1 - e^{-aT} z^{-1}}}$$

2. Impulse Invariant Transformation

Comparing the expression of $H(s)$ and $H(z)$ [i.e., equations (2.1) and (2.4)] we can say that,

$$\boxed{\frac{1}{s + p_i} \xrightarrow{\text{(is transformed to)}} \frac{1}{1 - e^{-p_i T} z^{-1}}} \quad \text{.....(2.5)}$$

2. Impulse Invariant Transformation

2.1 Relation Between Analog and Digital Frequency in Impulse Invariant Transformation

Let, Ω = Analog frequency in rad/second.

ω = Digital frequency in rad/sample.

$$\boxed{\text{Digital frequency, } \omega = \Omega T} \quad \text{or} \quad \boxed{\text{Analog frequency, } \Omega = \frac{\omega}{T}} \quad \dots (2.1.1)$$

2. Impulse Invariant Transformation

2.2 Useful Impulse Invariant Transformation

The following transformations are given without proof. The equation (2.2.2) can be used when the analog real poles has a multiplicity of m . The equations (2.2.3) and (2.2.4) can be used when the analog poles are complex conjugate.

$$\frac{1}{(s + p_i)^m} \longrightarrow \frac{(-1)^{m-1}}{(m-1)!} \frac{d^{m-1}}{dp_i^{m-1}} \frac{1}{1 - e^{-p_i T} z^{-1}} \quad \text{.....(2.2.2)}$$

$$\frac{(s + a)}{(s + a)^2 + b^2} \longrightarrow \frac{1 - e^{-aT} (\cos bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}} \quad \text{....(2.2.3)}$$

$$\frac{b}{(s + a)^2 + b^2} \longrightarrow \frac{e^{-aT} (\sin bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}} \quad \text{....(2.2.4)}$$

2. Impulse Invariant Transformation

Example 1

For the analog transfer function, $H(s) = \frac{2}{s^2 + 3s + 2}$, determine $H(z)$ using impulse invariant transformation if **(a)** $T = 1$ second and **(b)** $T = 0.1$ second.

2. Impulse Invariant Transformation

Solution

Given that, $H(s) = \frac{2}{s^2 + 3s + 2} = \frac{2}{(s + 1)(s + 2)}$

By partial fraction expansion technique we can write,

$$H(s) = \frac{2}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2}$$

$$A = \frac{2}{(\cancel{s + 1})(s + 2)} \times (\cancel{s + 1}) \Big|_{s = -1} = \frac{2}{-1 + 2} = 2$$

$$B = \frac{2}{(s + 1)(\cancel{s + 2})} \times (\cancel{s + 2}) \Big|_{s = -2} = \frac{2}{-2 + 1} = -2$$

$$\therefore H(s) = \frac{2}{s + 1} + \frac{-2}{s + 2}$$

2. Impulse Invariant Transformation

By impulse invariant transformation we know that,

$$\boxed{\frac{A_i}{s + p_i} \xrightarrow{\text{(is transformed to)}} \frac{A_i}{1 - e^{-p_i T} z^{-1}}}$$

$$\therefore H(z) = \frac{2}{1 - e^{-p_1 T} z^{-1}} + \frac{-2}{1 - e^{-p_2 T} z^{-1}} \quad \text{where } p_1 = 1 \quad \text{and } p_2 = 2$$

$$H(z) = \frac{2}{1 - e^{-T} z^{-1}} + \frac{-2}{1 - e^{-2T} z^{-1}}$$

(a) When T = 1 second

$$H(z) = \frac{2}{1 - e^{-1} z^{-1}} + \frac{-2}{1 - e^{-2} z^{-1}}$$

$$H(z) = \frac{2}{1 - 0.3679z^{-1}} + \frac{-2}{1 - 0.1353z^{-1}} = \frac{2(1 - 0.1353z^{-1}) - 2(1 - 0.3679z^{-1})}{(1 - 0.3679z^{-1})(1 - 0.1353z^{-1})}$$

$$= \frac{2 - 0.2706z^{-1} - 2 + 0.7358z^{-1}}{1 - 0.1353z^{-1} - 0.3679z^{-1} + 0.0498z^{-2}} = \frac{0.4652z^{-1}}{1 - 0.5032z^{-1} + 0.0498z^{-2}}$$

2. Impulse Invariant Transformation

(b) When T = 0.1 second

$$\begin{aligned} H(z) &= \frac{2}{1 - e^{-0.1} z^{-1}} + \frac{-2}{1 - e^{-0.2} z^{-1}} \\ &= \frac{2}{1 - 0.9048z^{-1}} + \frac{-2}{1 - 0.8187z^{-1}} = \frac{2(1 - 0.8187z^{-1}) - 2(1 - 0.9048z^{-1})}{(1 - 0.9048z^{-1})(1 - 0.8187z^{-1})} \\ &= \frac{2 - 1.6374z^{-1} - 2 + 1.8096z^{-1}}{1 - 0.8187z^{-1} - 0.9048z^{-1} + 0.7408z^{-2}} = \frac{0.1722z^{-1}}{1 - 1.7235z^{-1} + 0.7408z^{-2}} \end{aligned}$$

2. Impulse Invariant Transformation

Example 2: [Home Work]

Convert the analog filter with system transfer function,

$$H(s) = \frac{(s + 0.1)}{(s + 0.1)^2 + 9}$$

into a digital IIR filter by means of the impulse invariant method.

Note : Evaluate $\cos \theta$ by keeping calculator in radian mode.

3. Bilinear Transformation

The ***bilinear transformation*** is a conformal mapping that transforms the imaginary axis of s-plane into the unit circle in the z-plane only once, thus avoiding aliasing of frequency components. In this mapping all points in the left half of s-plane are mapped inside the unit circle in the z-plane and all points in the right half of s-plane are mapped outside the unit circle in the z-plane.

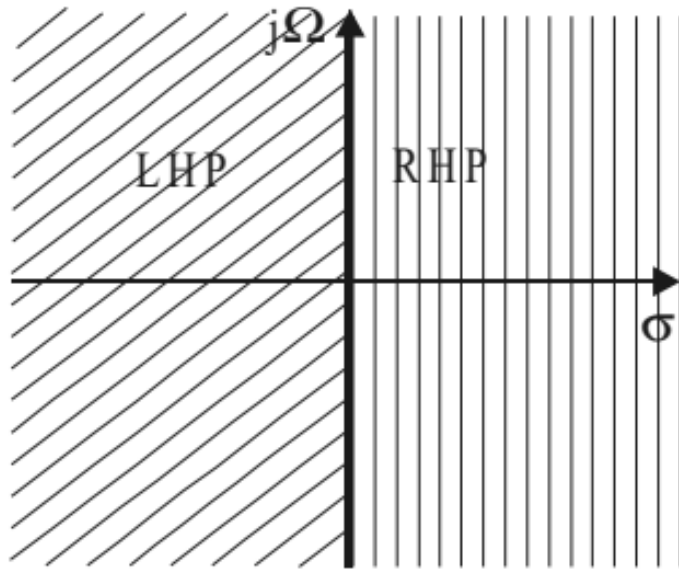


Fig 1a : s-plane.

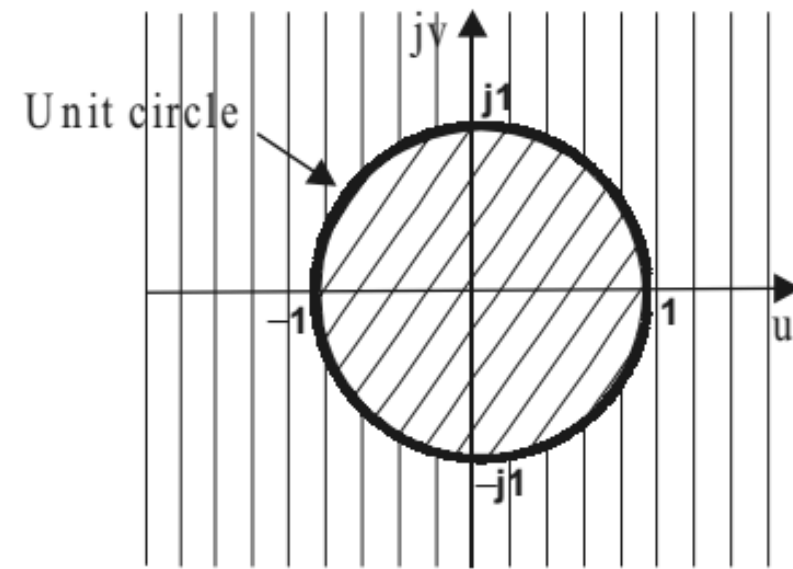


Fig 1b : z-plane.

Fig 1 : Mapping of s-plane into z-plane in bilinear transformation.

3. Bilinear Transformation

The bilinear transformation can be linked to the trapezoidal formula for numerical integration. Any analog system is governed by a differential equation in time domain. Consider the first order differential equation of an analog system as shown in equation (3.1).

$$\text{Let, } \frac{dy(t)}{dt} = x(t) \quad \text{.....(3.1)}$$

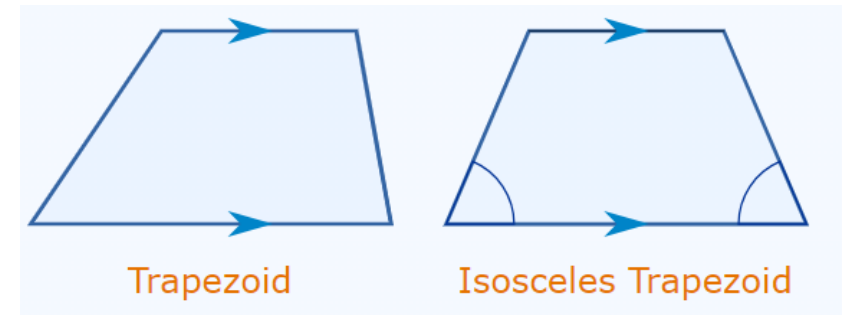
3. Bilinear Transformation

On integrating both sides of equation (3.1) we get,

$$\int_{(n-1)T}^{nT} \frac{dy(t)}{dt} dt = \int_{(n-1)T}^{nT} x(t) dt$$
$$[y(t)]_{(n-1)T}^{nT} = \int_{(n-1)T}^{nT} x(t) dt$$
$$y(nT) - y((n-1)T) = \int_{(n-1)T}^{nT} x(t) dt \dots\dots(3.2)$$

The trapezoidal rule when integration is approximated by two trapezoids is,

$$\int_a^b f(x) dx = \frac{b-a}{2} [f(a) + f(b)]$$



The integral on the right side of equation (3.2) can be approximated by the trapezoidal rule, so that,

$$y(nT) - y[(n-1)T] = \left(\frac{T}{2}\right) [x(nT) + x((n-1)T)] \dots\dots(3.3)$$

3. Bilinear Transformation

For discrete time system, the equation (3.3) can be written as,

$$y(n) - y(n-1) = \frac{T}{2} [x(n) + x(n-1)] \dots\dots(3.4)$$

On taking $\mathcal{Z}$ -transform of equation (3.4) we get,

$$Y(z) - z^{-1} Y(z) = \frac{T}{2} [X(z) + z^{-1} X(z)]$$

$$[1 - z^{-1}] Y(z) = \frac{T}{2} [1 + z^{-1}] X(z)$$

$$\frac{2(1 - z^{-1})}{T(1 + z^{-1})} Y(z) = X(z) \dots\dots(3.5)$$

3. Bilinear Transformation

On taking Laplace transform of equation (3.1) we get,

$$s Y(s) = X(s) \dots\dots(3.6)$$

On comparing equations (3.5) and (3.6) we can say that,

$$s Y(s) \xrightarrow{\text{(is transformed to)}} \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} Y(z) \dots\dots(3.7)$$

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

$$Y(s) = Y(z)$$

by bilinear transformation, where T is the sampling time period.

Hence in the s-domain transfer function, if "s" is substituted by the term $\frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$ the resulting transfer function will be z-domain transfer function.

3. Bilinear Transformation

3.1 Relation Between Analog and Digital Frequency in Bilinear Transformation

Let, $s = j\Omega$ be points on imaginary axis and the corresponding points on the z-plane on unit circle are given by $z = e^{j\omega}$. For bilinear transformation,

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \dots\dots(3.1.1)$$

Put, $s = j\Omega$ and $z = e^{j\omega}$ in equation (3.1.1)

$$\therefore j\Omega = \frac{2}{T} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = \frac{2}{T} \frac{\left(e^{\frac{j\omega}{2}} e^{\frac{-j\omega}{2}} - e^{-j\omega} \right)}{\left(e^{\frac{j\omega}{2}} e^{\frac{-j\omega}{2}} + e^{-j\omega} \right)}$$

$$e^{j\theta} e^{-j\theta} = 1$$

3. Bilinear Transformation

$$j\Omega = \frac{2}{T} \frac{e^{\frac{-j\omega}{2}} \left(e^{\frac{j\omega}{2}} - e^{\frac{-j\omega}{2}} \right)}{e^{\frac{-j\omega}{2}} \left(e^{\frac{j\omega}{2}} + e^{\frac{-j\omega}{2}} \right)} = \frac{2}{T} \frac{2j \sin \frac{\omega}{2}}{2 \cos \frac{\omega}{2}}$$

$$\therefore \Omega = \frac{2}{T} \frac{\sin \frac{\omega}{2}}{\cos \frac{\omega}{2}} = \frac{2}{T} \tan \frac{\omega}{2}$$

$$\therefore \text{Analog frequency, } \Omega = \frac{2}{T} \tan \frac{\omega}{2} \text{(3.1.2)}$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

- $2j \sin \theta = e^{j\theta} - e^{-j\theta}$
- $2 \cos \theta = e^{j\theta} + e^{-j\theta}$
- $\tan = \sin \theta / \cos \theta$

3. Bilinear Transformation

The equation (3.1.2) relates the analog frequency, Ω and digital frequency, ω .

From equation (3.1.2) we get,

$$\frac{\Omega T}{2} = \tan \frac{\omega}{2} \quad \Rightarrow \quad \frac{\omega}{2} = \tan^{-1} \frac{\Omega T}{2}$$

$\therefore \text{Digital frequency, } \omega = 2 \tan^{-1} \frac{\Omega T}{2} \text{(3.1.3)}$

The equation (3.1.3) can be used to estimate the digital frequency ω for a given analog frequency, Ω . The equation (3.1.2) is used to calculate the analog frequency for a given digital frequency.

3. Bilinear Transformation

Example 3

For the analog transfer function, $H(s) = \frac{2}{s^2 + 3s + 2}$, determine $H(z)$ using bilinear transformation if
(a) $T = 1$ second and **(b)** $T = 0.1$ second.

3. Bilinear Transformation

Solution

Given that, $H(s) = \frac{2}{s^2 + 3s + 2}$

Put, $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$ in $H(s)$ to get $H(z)$.

$$\begin{aligned}\therefore H(z) &= \frac{2}{\left(\frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}\right)^2 + 3 \left(\frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}\right) + 2} \\ &= \frac{2}{\frac{4}{T^2} \frac{(1 - z^{-1})^2}{(1 + z^{-1})^2} + \frac{6}{T} \frac{(1 - z^{-1})}{(1 + z^{-1})} + 2}\end{aligned}$$

3. Bilinear Transformation

$$\begin{aligned}\therefore H(z) &= \frac{2}{4(1-z^{-1})^2 + 6T(1-z^{-1})(1+z^{-1}) + 2T^2(1+z^{-1})^2} \\ &= \frac{2T^2(1+z^{-1})^2}{4(1-z^{-1})^2 + 6T(1-z^{-2}) + 2T^2(1+z^{-1})^2}\end{aligned}$$

$$(a+b)(a-b) = a^2 - b^2$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

3. Bilinear Transformation

(a) T = 1 second

$$\begin{aligned}\therefore H(z) &= \frac{2(1+z^{-1})^2}{4(1-z^{-1})^2 + 6(1-z^{-2}) + 2(1+z^{-1})^2} \\&= \frac{2(1+2z^{-1}+z^{-2})}{4(1-2z^{-1}+z^{-2}) + 6(1-z^{-2}) + 2(1+2z^{-1}+z^{-2})} \\&= \frac{2+4z^{-1}+2z^{-2}}{12-4z^{-1}} = \frac{2+4z^{-1}+2z^{-2}}{12\left(1-\frac{4}{12}z^{-1}\right)} \\&= \frac{\frac{2}{12} + \frac{4}{12}z^{-1} + \frac{2}{12}z^{-2}}{1-\frac{4}{12}z^{-1}} = \frac{0.1667 + 0.3333z^{-1} + 0.1667z^{-2}}{1-0.3333z^{-1}}\end{aligned}$$

3. Bilinear Transformation

(b) T = 0.1 second

$$\begin{aligned} H(z) &= \frac{2 \times 0.1^2 (1 + z^{-1})^2}{4(1 - z^{-1})^2 + 6 \times 0.1(1 - z^{-2}) + 2 \times 0.1^2 (1 + z^{-1})^2} \\ &= \frac{0.02(1 + 2z^{-1} + z^{-2})}{4(1 - 2z^{-1} + z^{-2}) + 0.6(1 - z^{-2}) + 0.02(1 + 2z^{-1} + z^{-2})} \\ &= \frac{0.02 + 0.04z^{-1} + 0.02z^{-2}}{4.62 - 7.96z^{-1} + 3.42z^{-2}} \\ &= \frac{\frac{0.02}{4.62} + \frac{0.04}{4.62}z^{-1} + \frac{0.02}{4.62}z^{-2}}{1 - \frac{7.96}{4.62}z^{-1} + \frac{3.42}{4.62}z^{-2}} = \frac{0.0043 + 0.0087z^{-1} + 0.0043z^{-2}}{1 - 1.7229z^{-1} + 0.7403z^{-2}} \end{aligned}$$

3. Bilinear Transformation

Example 4: [Home Work]

Obtain $H(z)$ from $H(s)$ when $T = 1$ second and $H(s) = \frac{2s}{s^2 + 0.2s + 1}$ (using bilinear transformation)

Example 5: [Home Work]

Obtain $H(z)$ from $H(s)$ when $T = 1$ second, and $H(s) = \frac{s^3}{(s + 1)(s^2 + s + 1)}$ (using bilinear transformation)

Example 6: [Home Work]

Convert the analog filter with system function $H(s)$ into digital filter using bilinear transformation.

$$H(s) = \frac{s + 0.3}{(s + 0.3)^2 + 16} \quad ; \quad \text{Take } T = 0.5$$

References: Textbooks

1. *“Discrete-Time Signal Processing”, International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*